Homework 03 - Tune and evaluate models

INFO 4940/5940 - Fall 2025

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2025-09-22

Setup

Load packages and data:

```
library(yaml)
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.2
✔ ggplot2   4.0.0     ✔ tibble    3.3.0
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.1.0
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(tidymodels)
```

```
── Attaching packages ────────────────────────────────────── tidymodels 1.4.1 ──
✔ broom        1.0.10     ✔ rsample      1.3.1
✔ dials        1.4.2      ✔ tailor       0.1.0
✔ infer        1.0.9      ✔ tune         2.0.0
✔ modeldata    1.5.1      ✔ workflows    1.3.0
✔ parsnip      1.3.3      ✔ workflowsets 1.1.1
✔ recipes      1.3.1      ✔ yardstick    1.3.2
── Conflicts ───────────────────────────────────────── tidymodels_conflicts() ──
```

```
──
✖ scales::discard() masks purrr::discard()
✖ dplyr::filter()   masks stats::filter()
✖ recipes::fixed()  masks stringr::fixed()
✖ dplyr::lag()      masks stats::lag()
✖ yardstick::spec() masks readr::spec()
✖ recipes::step()   masks stats::step()
```

```r
library(arrow)
```

```

Attaching package: 'arrow'

The following object is masked from 'package:dials':

    buffer

The following object is masked from 'package:lubridate':

    duration

The following object is masked from 'package:utils':

    timestamp
```

```r
library(themis)

set.seed(123)
```

Exercise 1

```r
# Metrics Identified for Performance

# Since this is a classification problem predicting marijuana legalization
attitudes, we need metrics that

# a) Handle class imbalances appropriately
# b) Evaluate performance on BOTH classes equally
# c) Are interpretable for potential outside stakeholders
```

```r
# 1. Sensitivity (Recall/True Positive Rate)
# Reason: The proportion of 'shoudld not be legal' responses correctly
identified, plus it's important for understanding pro-legalization
sentiment(s)
```

```
# 2. Specificity (True Negative Rate)
# Reason: The proportion of 'should not be legal' responses correctly
identified & it is important for understanding opposition sentiment

# 3. Balanced Accuracy (Sensitivity + Specificty) / 2
# Reason: Addresses class imbalance by weighing both classes equally. It also
is a single metric that balance performanceo n BOTH classes.

# 4. ROC AUC (Area under ROC Curve)
# Reason: It measures discriminative ability across all probability thresholds
and it is less sensitive to class imbalances than a regular accuracy

# 5. J-Index (also known as Youden's Index): Sensitivity + Specificity - 1
# Reason: It maximizes the difference between true positive rate and false
positive rate and it dirrectly related to balanced accuracy (same ranking)

# 6. Brier Score
# Reason: It measures calibration (how well predicted probabilities match
actual outcomes) and will usually offer lower scores. Lower scores indicate
better probability estimates.
```

Exercise 2

```
gss_data <- read_feather("data/gss.feather")

glimpse(gss_data)
```

```
Rows: 1,591
Columns: 25
$ id       <int> 2, 3, 6, 8, 10, 11, 12, 13, 14, 15, 16, 23, 26, 28, 29, 30,
3…
$ age      <dbl> 72, 25, 36, 76, 84, 64, 49, 29, 50, 38, 28, 32, 24, 29, 47,
5…
$ colath   <fct> NA, "yes, allowed to teach", "not allowed", NA, NA, "yes,
all…
$ colmslm  <fct> NA, "not allowed", "not allowed", NA, NA, "not allowed", NA,
…
$ degree   <ord> high school, bachelor's, bachelor's, high school, associate/
j…
$ fear     <fct> NA, yes, no, NA, NA, no, NA, NA, yes, no, NA, NA, NA, NA,
NA,…
$ grass    <fct> should be legal, should be legal, should be legal, should be
…
$ gunlaw   <fct> NA, NA, favor, NA, NA, favor, NA, NA, favor, favor, NA, NA,
N…
```

```
$ happy    <ord> pretty happy, pretty happy, not too happy, pretty happy,
pret…
$ health   <fct> excellent, fair, fair, poor, poor, excellent, good, good,
goo…
$ hispanic <fct> "not hispanic", "another hispanic, latino, or spanish
origin"…
$ hrs1     <dbl> NA, 33, 0, NA, NA, 50, 36, NA, 89, 55, 40, 38, 30, 45, NA,
40…
$ income16 <ord> "$10,000 to $12,499", "$30,000 to $34,999", "$50,000 to
$59,9…
$ letdie1  <fct> NA, NA, NA, NA, no, NA, no, yes, NA, NA, yes, no, yes, NA,
NA…
$ owngun   <fct> NA, no, no, NA, NA, no, NA, NA, no, no, NA, NA, NA, NA, NA,
N…
$ partyid  <ord> "strong democrat", "independent, close to democrat", "not
ver…
$ polviews <ord> "moderate, middle of the road", NA, "moderate, middle of the
…
$ pray     <fct> several times a day, less than once a week, several times a
w…
$ pres20   <fct> biden, biden, biden, biden, NA, biden, trump, NA, biden, NA,
…
$ race     <fct> NA, white, other, white, white, white, white, other, black,
w…
$ region   <fct> middle atlantic, middle atlantic, middle atlantic, middle
atl…
$ sex      <fct> female, female, male, female, male, male, female, female,
fem…
$ sexfreq  <ord> not at all, NA, once a month, not at all, NA, once or twice,
…
$ wordsum  <dbl> 4, NA, NA, 8, 7, NA, 8, 9, NA, NA, 6, 4, 7, 10, 1, 8, NA, 1,
…
$ wrkstat  <fct> "retired", "working part time", "working full time",
"retired…
```

```
# Checking outcome variable distro
gss_data |>
  count(grass) |>
  mutate(proportion = n / sum(n))
```

```
# A tibble: 2 × 3
  grass                   n proportion
  <fct>               <int>      <dbl>
1 should be legal      1105      0.695
2 should not be legal   486      0.305
```

```r
summary(gss_data)
```

```
       id              age                          colath
 Min.   :   2.0   Min.   :18.00   yes, allowed to teach:473
 1st Qu.: 501.0   1st Qu.:36.00   not allowed          :282
 Median : 973.0   Median :53.00   NA's                 :836
 Mean   : 992.9   Mean   :52.06
 3rd Qu.:1492.5   3rd Qu.:67.00
 Max.   :1990.0   Max.   :89.00
                  NA's   :69
                  colmslm                          degree       fear
 yes, allowed to teach:204   less than high school   :207   yes :270
 not allowed          :546   high school             :793   no  :503
 NA's                 :841   associate/junior college:136   NA's:818
                             bachelor's              :282
                             graduate                :173


                 grass          gunlaw              happy           health
 should be legal    :1105   favor :517   very happy   :334   excellent:262
 should not be legal: 486   oppose:243   pretty happy :854   good     :767
                            NA's  :831   not too happy:389   fair     :449
                                         NA's         : 14   poor     :111
                                                             NA's     :  2



                                       hispanic          hrs1
 not hispanic                              :1332   Min.   : 0.0
 mexican, mexican american, chicano/a      : 116   1st Qu.:35.0
 puerto rican                              :  33   Median :40.0
 cuban                                     :  17   Mean   :40.1
 another hispanic, latino, or spanish origin:  78   3rd Qu.:47.0
 NA's                                      :  15   Max.   :89.0
                                                   NA's   :844
                income16    letdie1          owngun
 $60,000 to $74,999 :128   yes : 265   yes    :245
 $50,000 to $59,999 :117   no  : 115   no     :507
 $90,000 to $109,999:109   NA's:1211   refused: 27
 $170,000 or over   :102               NA's   :812
 $75,000 to $89,999 : 97
 (Other)            :859
 NA's               :179
                              partyid                               polviews
 independent (neither, no response):348   moderate, middle of the road:537
 strong democrat                   :261   conservative                :222
 not very strong democrat          :219   slightly conservative       :199
 strong republican                 :197   slightly liberal            :194
```

```
 independent, close to democrat    :179   liberal                     :171  
 (Other)                           :365   (Other)                     :169  
 NA's                              : 22   NA's                        : 99  
                   pray                                pres20        race     
 several times a day  :587   biden                     :598   white:1012  
 once a day           :349   trump                     :422   black: 355  
 several times a week :164   other candidate           : 28   other: 191  
 once a week          : 88   didn't vote for president:  5   NA's :  33  
 less than once a week:139   NA's                      :538               
 never                :238                                                
 NA's                 : 26                                                
               region          sex                         sexfreq   
 south atlantic    :349   male  :728   not at all          :411  
 east north central:288   female:861   2 or 3 times a month:168  
 middle atlantic   :196   NA's  :  2   2 or 3 times a week :157  
 pacific           :189                once a month        :144  
 west south central:163                about once a week   :134  
 east south central:151                (Other)             :168  
 (Other)           :255                NA's                :409  
    wordsum                                            wrkstat   
 Min.   : 0.000   working full time                        :602  
 1st Qu.: 4.000   retired                                  :438  
 Median : 6.000   keeping house                            :171  
 Mean   : 5.489   working part time                        :150  
 3rd Qu.: 7.000   unemployed, laid off, looking for work: 86  
 Max.   :10.000   (Other)                                  :141  
 NA's   :806      NA's                                     :  3  
```

```r
# checking missing patterns

gss_data |>
  summarise(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(
    everything(),
    names_to = "variable",
    values_to = "missing_count"
  ) |>
  mutate(missing_prop = missing_count / nrow(gss_data)) |>
  arrange(desc(missing_count))
```

```
# A tibble: 25 × 3
   variable missing_count missing_prop
   <chr>            <int>        <dbl>
 1 letdie1           1211        0.761
 2 hrs1               844        0.530
 3 colmslm            841        0.529
```

```
 4 colath              836        0.525
 5 gunlaw              831        0.522
 6 fear                818        0.514
 7 owngun              812        0.510
 8 wordsum             806        0.507
 9 pres20              538        0.338
10 sexfreq             409        0.257
# ℹ 15 more rows
```

```
gss_data |>
  select(grass, gunlaw, partyid, region, sex, fear) |>
  summary()
```

```
                 grass          gunlaw
 should be legal    :1105   favor :517
 should not be legal: 486   oppose:243
                            NA's  :831




                               partyid                   region
sex
 independent (neither, no response):348   south atlantic    :349
male  :728
 strong democrat                   :261   east north central:288
female:861
 not very strong democrat          :219   middle atlantic   :196   NA's  :
2
 strong republican                 :197
pacific           :189
 independent, close to democrat    :179   west south
central:163
 (Other)                           :365   east south
central:151
 NA's                              : 22
(Other)           :255
   fear
 yes :270
 no  :503
 NA's:818
```

```r
# Class imbalance?

gss_data |>
  count(grass) |>
  mutate(proportion = n / sum(n)) |>
  mutate(ratio = paste0(round(proportion * 100, 1), "%"))
```

```
# A tibble: 2 × 4
  grass                    n proportion ratio
  <fct>                <int>      <dbl> <chr>
1 should be legal       1105      0.695 69.5%
2 should not be legal    486      0.305 30.5%
```

```r
# 69.5% favor (1105 respondents)
# 30.5% oppose (486 respondents)
# That is ~2.3:1 in favor
```

```r
# Missing Data?

gss_data |>
  summarise(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(
    everything(),
    names_to = "variable",
    values_to = "missing_count"
  ) |>
  mutate(missing_prop = missing_count / nrow(gss_data)) |>
  arrange(desc(missing_count)) |>
  filter(missing_prop > 0) |> # Only show variables with missing data
  print(n = 15)
```

```
# A tibble: 21 × 3
   variable missing_count missing_prop
   <chr>            <int>        <dbl>
 1 letdie1           1211       0.761
 2 hrs1               844       0.530
 3 colmslm            841       0.529
 4 colath             836       0.525
 5 gunlaw             831       0.522
 6 fear               818       0.514
 7 owngun             812       0.510
 8 wordsum            806       0.507
 9 pres20             538       0.338
10 sexfreq            409       0.257
```

```
11 income16          179       0.113
12 polviews           99       0.0622
13 age                69       0.0434
14 race               33       0.0207
15 pray               26       0.0163
# ℹ 6 more rows
```

Analysis: There are many missing data across varables.

over 50% missing: letdie1, hrs1, gunlaw) # suggests these questions weren't asked to all respondents. # Cant just drop rows with missing data, as we would lose almost all of the dataset # Going to use mean imputation for numeric variables & new category for categorical analysis

```
# gunlaw vs grass

if (sum(!is.na(gss_data$gunlaw)) > 0) {
  gss_data |>
    filter(!is.na(gunlaw)) |>
    count(grass, gunlaw) |>
    group_by(gunlaw) |>
    mutate(prop = n / sum(n)) |>
    print()
}
```

```
# A tibble: 4 × 4
# Groups:   gunlaw [2]
  grass               gunlaw     n  prop
  <fct>               <fct>  <int> <dbl>
1 should be legal     favor    368 0.712
2 should be legal     oppose   178 0.733
3 should not be legal favor    149 0.288
4 should not be legal oppose    65 0.267
```

Analysis: Surprising lack of strong relationship.

Unlike expected, gun attributes do not strongly predict drug attitudes. Government framework might not apply uniformly across policy domains. It also shows that these indicators are not great predictors

```
# partyid vs. grass

gss_data |>
  filter(!is.na(partyid)) |>
  count(grass, partyid) |>
  group_by(partyid) |>
```

```
  mutate(prop = round(n / sum(n), 3)) |>
  arrange(partyid) |>
  print(n = 20)
```

```
# A tibble: 16 × 4
# Groups:   partyid [8]
   grass              partyid                                n  prop
   <fct>              <ord>                              <int> <dbl>
 1 should be legal     strong democrat                      204 0.782
 2 should not be legal strong democrat                       57 0.218
 3 should be legal     not very strong democrat             169 0.772
 4 should not be legal not very strong democrat              50 0.228
 5 should be legal     independent, close to democrat       146 0.816
 6 should not be legal independent, close to democrat        33 0.184
 7 should be legal     independent (neither, no response)   251 0.721
 8 should not be legal independent (neither, no response)    97 0.279
 9 should be legal     independent, close to republican      97 0.647
10 should not be legal independent, close to republican      53 0.353
11 should be legal     not very strong republican           105 0.587
12 should not be legal not very strong republican            74 0.413
13 should be legal     strong republican                     91 0.462
14 should not be legal strong republican                    106 0.538
15 should be legal     other party                           25 0.694
16 should not be legal other party                           11 0.306
```

Analysis: There is a strong political predictive pattern with a clear ideological gradient.

Theres a clear ideological spectrum from Strong Democrat (78%) to Strong Republican (46%). That's a 32% point spread. It will definitely be one of our strongest predictors

```
# region vs grass

gss_data |>
  count(grass, region) |>
  group_by(region) |>
  mutate(prop = round(n / sum(n), 3)) |>
  arrange(desc(prop)) |>
  print(n = 20)
```

```
# A tibble: 18 × 4
# Groups:   region [9]
   grass              region                 n  prop
   <fct>              <fct>              <int> <dbl>
 1 should be legal     mountain              83 0.735
```

```
 2 should be legal     middle atlantic      143 0.73
 3 should be legal     new england           39 0.722
 4 should be legal     east north central   208 0.722
 5 should be legal     pacific              132 0.698
 6 should be legal     south atlantic       243 0.696
 7 should be legal     east south central   102 0.675
 8 should be legal     west north central    55 0.625
 9 should be legal     west south central   100 0.613
10 should not be legal west south central    63 0.387
11 should not be legal west north central    33 0.375
12 should not be legal east south central    49 0.325
13 should not be legal south atlantic       106 0.304
14 should not be legal pacific               57 0.302
15 should not be legal new england           15 0.278
16 should not be legal east north central    80 0.278
17 should not be legal middle atlantic       53 0.27
18 should not be legal mountain              30 0.265
```

Analysis: Regional variation but less dramatic than political party. Still will be a useful predictor but secondary to political party.

```
# Sex vs Grass?

gss_data |>
  filter(!is.na(sex)) |>
  count(grass, sex) |>
  group_by(sex) |>
  mutate(prop = round(n / sum(n), 3)) |>
  print()
```

```
# A tibble: 4 × 4
# Groups:   sex [2]
  grass               sex        n  prop
  <fct>               <fct>  <int> <dbl>
1 should be legal     male     527 0.724
2 should be legal     female   576 0.669
3 should not be legal male     201 0.276
4 should not be legal female   285 0.331
```

Analysis: Sex shows a moderate but consistent pattern; men are more supportive than women by ~5.5 percentage points. Meaningful, yes, but still much smaller than age or political party effects.

```r
# Age?
gss_data |>
  filter(!is.na(age)) |>
  mutate(
    age_group = case_when(
      age < 30 ~ "Under 30",
      age < 45 ~ "30-44",
      age < 60 ~ "45-59",
      age < 75 ~ "60-74",
      TRUE ~ "75+"
    )
  ) |>
  count(grass, age_group) |>
  group_by(age_group) |>
  mutate(prop = round(n / sum(n), 3)) |>
  arrange(age_group) |>
  print()
```

```
# A tibble: 10 × 4
# Groups:   age_group [5]
   grass               age_group     n  prop
   <fct>               <chr>     <int> <dbl>
 1 should be legal     30-44       283 0.761
 2 should not be legal 30-44        89 0.239
 3 should be legal     45-59       241 0.701
 4 should not be legal 45-59       103 0.299
 5 should be legal     60-74       278 0.654
 6 should not be legal 60-74       147 0.346
 7 should be legal     75+          72 0.404
 8 should not be legal 75+         106 0.596
 9 should be legal     Under 30    179 0.882
10 should not be legal Under 30     24 0.118
```

Analysis: Wow, age seems like a pretty powerful predictor. There's a 48% point spread from Under 30 to 75+, which is huge. Maybe there's generational replacement driving the shift. Older cohorts and the 'war on drugs' vs. younger cohorts with different cultural experiences.

```r
# Particioning

set.seed(123)

gss_split <- initial_split(gss_data, prop = 0.8, strata = grass)
gss_train <- training(gss_split)
gss_test <- testing(gss_split)
```

```r
# Verification

gss_train |>
  count(grass) |>
  mutate(prop = round(n / sum(n), 3))
```

```
# A tibble: 2 × 3
  grass                   n  prop
  <fct>               <int> <dbl>
1 should be legal       884 0.695
2 should not be legal   388 0.305
```

```r
gss_test |>
  count(grass) |>
  mutate(prop = round(n / sum(n), 3))
```

```
# A tibble: 2 × 3
  grass                   n  prop
  <fct>               <int> <dbl>
1 should be legal       221 0.693
2 should not be legal    98 0.307
```

Particioning Strategy:

1. 80/20 split: gives us 1,272 training & 319 test, which is large enough for training complex models and sufficient for the test set evaluation. # 2. Stratified sampling: because we have a class imbalance, stratification makes sure both training and test sets have nearly identical class distributions. This will avoid bias from chance imbalances. # 3. Set seed as required!

Exercise 3

```r
# Cross-valudcation folds
set.seed(123)

gss_folds <- vfold_cv(gss_train, v = 10, strata = grass)
```

Justification: Standard practice, which is good for bias-variance tradeoff. Stratifying fixes class imbalance.

```r
# verify

gss_folds$splits[[1]] |>
  analysis() |>
  count(grass) |>
  mutate(prop = round(n / sum(n), 3))
```

```
# A tibble: 2 × 3
  grass                     n  prop
  <fct>                 <int> <dbl>
1 should be legal         795 0.695
2 should not be legal     349 0.305
```

Analysis: Perfect execution.

```r
null_spec <- logistic_reg() |>
  set_engine("glm") |>
  set_mode("classification")

# recipe!
null_recipe <- recipe(grass ~ 1, data = gss_train)

# workflow
null_workflow <- workflow() |>
  add_model(null_spec) |>
  add_recipe(null_recipe)

# Evaluate null model
null_results <- null_workflow |>
  fit_resamples(
    resamples = gss_folds,
    metrics = metric_set(
      accuracy,
      sensitivity,
      specificity,
```

```
      roc_auc,
      bal_accuracy
    )
  )


null_metrics <- collect_metrics(null_results)
print("=== NULL MODEL PERFORMANCE ===")
```

```
[1] "=== NULL MODEL PERFORMANCE ==="
```

```
null_metrics
```

```
# A tibble: 5 × 6
  .metric      .estimator  mean     n  std_err .config
  <chr>        <chr>      <dbl> <int>    <dbl> <chr>
1 accuracy     binary     0.695    10 0.000676 pre0_mod0_post0
2 bal_accuracy binary     0.5      10 0        pre0_mod0_post0
3 roc_auc      binary     0.5      10 0        pre0_mod0_post0
4 sensitivity  binary     1        10 0        pre0_mod0_post0
5 specificity  binary     0        10 0        pre0_mod0_post0
```

Analysis: The intercept only baseline achieves a 69.5% accuracy because 69.5% of respondents favor legalization. But it also has a TPR = 1.00 & TNR = 0.00, which yields a balanced accuracy of .50 and ROC AUC of .50. It is a naive accuracy misleading under class imbalance.

We must prioritize balanced accuracy, sensitivity, specificity, ROC AUC, and at a later point calibration metrics.

Exercise 4

```
ex4_metrics <- metric_set(
  bal_accuracy, # primary under imbalance
  sensitivity,
  specificity,
  roc_auc, # threshold-free separability
  brier_class, # AML4TD Ch.15: probability calibration quality
  j_index # AML4TD Ch.11–12: threshold selection signal
)
```

```r
ex4_base_rec <-
  recipe(grass ~ ., data = gss_train) |>
  update_role(id, new_role = "id") |> # drop this line if no 'id'
  step_zv(all_predictors()) |>
  step_impute_median(all_numeric(), -all_outcomes()) |>
  step_impute_mode(all_nominal(), -all_outcomes()) |>
  step_dummy(all_nominal(), -all_outcomes(), one_hot = TRUE) |>
  step_normalize(all_numeric(), -all_outcomes())

# R4DS inspect: see what the model will actually get
prep_ex4 <- prep(ex4_base_rec)
juice(prep_ex4) |> glimpse()
```

```
Rows: 1,272
Columns: 115
$ id
<dbl> …
$ age
<dbl> …
$ hrs1
<dbl> …
$ wordsum
<dbl> …
$ grass
<fct> …
$ colath_yes..allowed.to.teach
<dbl> …
$ colath_not.allowed
<dbl> …
$ colmslm_yes..allowed.to.teach
<dbl> …
$ colmslm_not.allowed
<dbl> …
$ degree_1
<dbl> …
$ degree_2
<dbl> …
$ degree_3
<dbl> …
$ degree_4
<dbl> …
$ degree_5
<dbl> …
$ fear_yes
<dbl> …
$ fear_no
<dbl> …
```

```
$ gunlaw_favor
<dbl> …
$ gunlaw_oppose
<dbl> …
$ happy_1
<dbl> …
$ happy_2
<dbl> …
$ happy_3
<dbl> …
$ health_excellent
<dbl> …
$ health_good
<dbl> …
$ health_fair
<dbl> …
$ health_poor
<dbl> …
$ hispanic_not.hispanic
<dbl> …
$ hispanic_mexican..mexican.american..chicano.a
<dbl> …
$ hispanic_puerto.rican
<dbl> …
$ hispanic_cuban
<dbl> …
$ hispanic_another.hispanic..latino..or.spanish.origin
<dbl> …
$ income16_01
<dbl> …
$ income16_02
<dbl> …
$ income16_03
<dbl> …
$ income16_04
<dbl> …
$ income16_05
<dbl> …
$ income16_06
<dbl> …
$ income16_07
<dbl> …
$ income16_08
<dbl> …
$ income16_09
<dbl> …
$ income16_10
<dbl> …
```

```
$ income16_11
<dbl> …
$ income16_12
<dbl> …
$ income16_13
<dbl> …
$ income16_14
<dbl> …
$ income16_15
<dbl> …
$ income16_16
<dbl> …
$ income16_17
<dbl> …
$ income16_18
<dbl> …
$ income16_19
<dbl> …
$ income16_20
<dbl> …
$ income16_21
<dbl> …
$ income16_22
<dbl> …
$ income16_23
<dbl> …
$ income16_24
<dbl> …
$ income16_25
<dbl> …
$ income16_26
<dbl> …
$ letdie1_yes
<dbl> …
$ letdie1_no
<dbl> …
$ owngun_yes
<dbl> …
$ owngun_no
<dbl> …
$ owngun_refused
<dbl> …
$ partyid_1
<dbl> …
$ partyid_2
<dbl> …
$ partyid_3
<dbl> …
```

```
$ partyid_4
<dbl> …
$ partyid_5
<dbl> …
$ partyid_6
<dbl> …
$ partyid_7
<dbl> …
$ partyid_8
<dbl> …
$ polviews_1
<dbl> …
$ polviews_2
<dbl> …
$ polviews_3
<dbl> …
$ polviews_4
<dbl> …
$ polviews_5
<dbl> …
$ polviews_6
<dbl> …
$ polviews_7
<dbl> …
$ pray_several.times.a.day
<dbl> …
$ pray_once.a.day
<dbl> …
$ pray_several.times.a.week
<dbl> …
$ pray_once.a.week
<dbl> …
$ pray_less.than.once.a.week
<dbl> …
$ pray_never
<dbl> …
$ pres20_biden
<dbl> …
$ pres20_trump
<dbl> …
$ pres20_other.candidate
<dbl> …
$ pres20_didn.t.vote.for.president
<dbl> …
$ race_white
<dbl> …
$ race_black
<dbl> …
```

```
$ race_other
<dbl> …
$ region_new.england
<dbl> …
$ region_middle.atlantic
<dbl> …
$ region_east.north.central
<dbl> …
$ region_west.north.central
<dbl> …
$ region_south.atlantic
<dbl> …
$ region_east.south.central
<dbl> …
$ region_west.south.central
<dbl> …
$ region_mountain
<dbl> …
$ region_pacific
<dbl> …
$ sex_male
<dbl> …
$ sex_female
<dbl> …
$ sexfreq_1
<dbl> …
$ sexfreq_2
<dbl> …
$ sexfreq_3
<dbl> …
$ sexfreq_4
<dbl> …
$ sexfreq_5
<dbl> …
$ sexfreq_6
<dbl> …
$ sexfreq_7
<dbl> …
$ wrkstat_working.full.time
<dbl> …
$ wrkstat_working.part.time
<dbl> …
$
wrkstat_with.a.job..but.not.at.work.because.of.temporary.illness..vacation..strike
<dbl> …
$ wrkstat_unemployed..laid.off..looking.for.work
<dbl> …
$ wrkstat_retired
```

```
<dbl> …
$ wrkstat_in.school
<dbl> …
$ wrkstat_keeping.house
<dbl> …
$ wrkstat_other
<dbl> …
```

```
ex4_down_rec <- ex4_base_rec |> themis::step_downsample(grass)
```

```
lasso_spec <-
  logistic_reg(penalty = tune(), mixture = 1) |>
  set_engine("glmnet") |>
  set_mode("classification")

pen_grid <- grid_regular(penalty(), levels = 30) # log-spaced by default
```

```
wf_orig <- workflow() |> add_model(lasso_spec) |> add_recipe(ex4_base_rec)
wf_down <- workflow() |> add_model(lasso_spec) |> add_recipe(ex4_down_rec)

ctrl <- control_resamples(save_pred = TRUE)

set.seed(123)
tune_orig <- tune_grid(
  wf_orig,
  resamples = gss_folds,
  grid = pen_grid,
  metrics = ex4_metrics,
  control = ctrl
)

set.seed(123)
tune_down <- tune_grid(
  wf_down,
  resamples = gss_folds,
  grid = pen_grid,
  metrics = ex4_metrics,
  control = ctrl
)
```

```
orig_tbl <- collect_metrics(tune_orig) |> mutate(set = "original")
down_tbl <- collect_metrics(tune_down) |> mutate(set = "downsampled")
```

```r
# quick look at primary metric
show_best(tune_orig, metric = "bal_accuracy", n = 5)
```

```
# A tibble: 5 × 7
   penalty .metric      .estimator  mean     n std_err .config
     <dbl> <chr>        <chr>      <dbl> <int>   <dbl> <chr>
1 3.86e- 3 bal_accuracy binary     0.665    10  0.0141 pre0_mod23_post0
2 7.88e- 4 bal_accuracy binary     0.662    10  0.0142 pre0_mod21_post0
3 3.56e- 4 bal_accuracy binary     0.661    10  0.0146 pre0_mod20_post0
4 1    e-10 bal_accuracy binary     0.661    10  0.0146 pre0_mod01_post0
5 2.21e-10 bal_accuracy binary     0.661    10  0.0146 pre0_mod02_post0
```

```r
show_best(tune_down, metric = "bal_accuracy", n = 5)
```

```
# A tibble: 5 × 7
   penalty .metric      .estimator  mean     n std_err .config
     <dbl> <chr>        <chr>      <dbl> <int>   <dbl> <chr>
1 0.0189   bal_accuracy binary     0.686    10  0.0149 pre0_mod25_post0
2 0.0418   bal_accuracy binary     0.684    10  0.0166 pre0_mod26_post0
3 0.00853  bal_accuracy binary     0.675    10  0.0153 pre0_mod24_post0
4 0.000788 bal_accuracy binary     0.662    10  0.0195 pre0_mod21_post0
5 0.00386  bal_accuracy binary     0.662    10  0.0188 pre0_mod23_post0
```

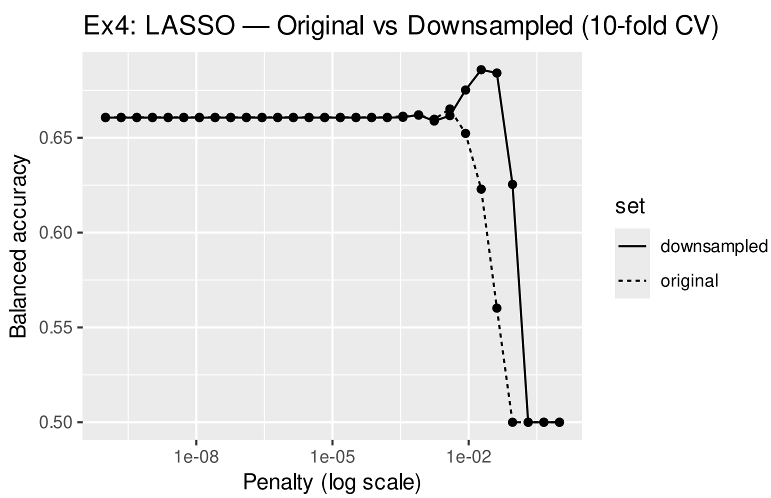
```r
# pick the best per approach for your comparison table
sel_orig <- select_best(tune_orig, metric = "bal_accuracy")
sel_down <- select_best(tune_down, metric = "bal_accuracy")

sel_orig
```

```
# A tibble: 1 × 2
  penalty .config
    <dbl> <chr>
1 0.00386 pre0_mod23_post0
```

```r
sel_down
```

```
# A tibble: 1 × 2
  penalty .config
    <dbl> <chr>
1  0.0189 pre0_mod25_post0
```

```r
bind_rows(
  orig_tbl |> filter(.metric == "bal_accuracy"),
  down_tbl |> filter(.metric == "bal_accuracy")
) |>
  ggplot(aes(penalty, mean, linetype = set)) +
  geom_line() +
  geom_point() +
  scale_x_log10() +
  labs(
    x = "Penalty (log scale)",
    y = "Balanced accuracy",
    title = "Ex4: LASSO — Original vs Downsampled (10-fold CV)"
  )
```



Exercise 5

Exercise 6

Exercise 7

Exercise 8

Exercise 9

Bonus (Battle Royale)

GAI self-reflection